

Parasitic diseases of the GI

Amoebiasis
Giardiasis
Balantidiasis

Opportunistic protozoa [Cyclosporiasis, Cryptosporidiosis, Isosporiosis]
Diphyllobothriasis

1

Parasitic diseases of the GI

Diseases	Causative species
Amoebiasis	Entamoeba histolytica
Giardiasis	Giardia lamblia
Balantidiasis	Balantidium coli
Cyclosporiasis Cryptosporidiosis Isosporiosis	Cyclospora cayetanensis Cryptosporidium parvum Isospora belli
Diphyllobothriasis	Diphyllobothrium latum

2

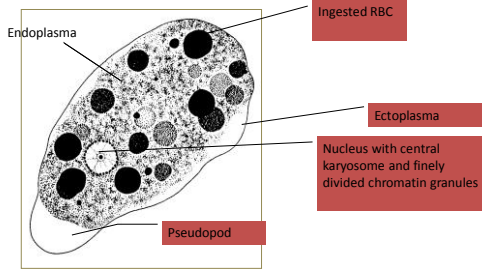
Parasitic protozoa of the GI: **classification**

The Phylum Protozoa is classified into four subdivisions according to the methods of locomotion.
◦The amoebae (Sarcodina) move by means of pseudopodia.
◦The flagellates (Mastigophora) typically move by long, whiplike flagellae.
◦The ciliates (Ciliata) are propelled by rows of cilia that beat with a synchronized wavelike motion.
◦The sporozoans (Sporozoa) lack specialized organelles of motility.

3

E. histolytica trophozoite

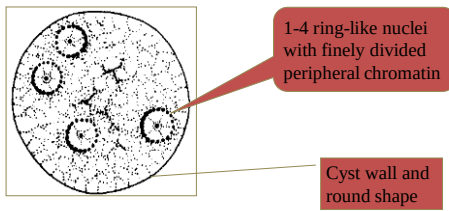
Morphology



4

Mature *E. histolytica* Cyst

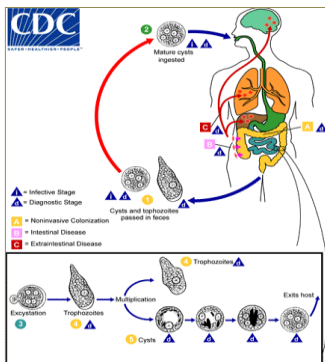
Morphology



5

Entamoeba histolytica

Life cycle



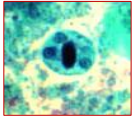
6

Entamoeba histolytica



Morphology & Laboratory Identification

- trophozoites range 12 to 30 microns in diameter; nucleus has an even distribution of peripheral chromatin and a small, compact, centrally located karyosome; cytoplasm is smooth and granular; inclusions, if present, are red blood cells;
- cysts range 10 to 20 microns in diameter and contains four nuclei when mature. Cigar-shaped chromatoid bars may be present in some cysts.



7

Amoeba in alimentary tract

- Entamoeba
 - *E. histolytica* (pathogenic)
 - *E. dispar* (non-pathogenic)
 - *E. coli* (big sister)
 - *E. hartmanni* (little brother)
 - *E. gingivalis* (oral)
- Endolimax nana (occasionally pathogenic)
- Iodamoeba butschlii

8

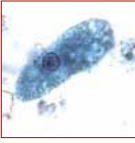
Entamoeba hartmanni



- Epidemiology - similar to *E. histolytica*
- Formerly called the “small race” of *Entamoeba histolytica*.
- Technologists must be able to differentiate this organism from *E. histolytica* because *E. hartmanni* is non-pathogenic.
- The difference lies in the sizes of the respective organisms. Trophozoites will measure less than 12 microns, while cysts will measure less than 10 microns.

9

Entamoeba coli



- Significance - this is a harmless commensal; must be differentiated from pathogens.
- Morphology - trophozoites range from 10 to 35 microns in diameter;
- Cysts range from 10 to 30 microns in diameter and contain 8 to 16 nuclei when mature; the nucleus exhibits an eccentric karyosome with irregular, coarse chromatin. The cytoplasm is heavily vacuolated, containing yeast, bacteria, and debris.

10

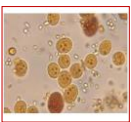
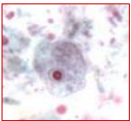
Entamoeba gingivalis



- Infective site - the mouth; the organism thrives in diseased gums, but is not considered a causal agent. It is destroyed in stomach if swallowed.
- Transmission - contact with fomites (drinking glasses, eating utensils, etc.); kissing.
- Morphology - resembles *E. histolytica*, but has no cyst stage. It is the only species which ingests leucocytes.

11

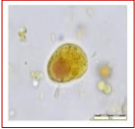
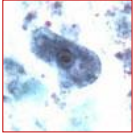
Endolimax nana



- Non-pathogenic; occasionally!
- Morphology - trophozoites range from 5 to 10 microns in diameter. The nucleus contains a large, blot-like karyosome; there is little or no peripheral chromatin.
- Cysts are usually sub-oval, measuring 4 to 6 by 6 to 10 microns.

12

Iodamoeba butschlii



- Non-pathogenic
- Morphology - the cyst is often called the "iodine cyst" due to the presence of a large glycogen vacuole which stains dark brown with iodine.

13

Entamoeba histolytica _ epidemiology

- **Prevalence** of amebic infection varies with level of sanitation and generally higher in tropics and subtropics than in temperate climates.
- Worldwide prevalence is about 10% to 50%
- **Cyst passers** are important source of infection
- **Entamoeba histolytica** is the second leading cause of mortality due to parasitic disease in humans. (The first being malaria). Amebiasis is the cause of an estimated 50,000-100,000 deaths each year.

14

Entamoeba histolytica _ transmission

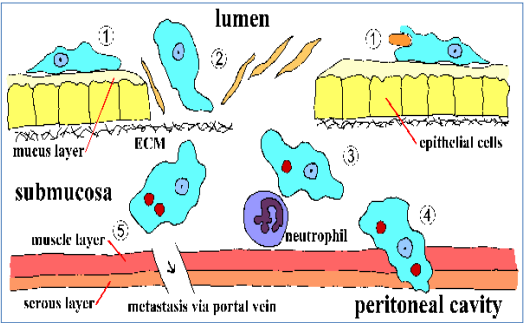
- Feco-oral: via cysts; i.e., food or drink contaminated with feces containing E. histolytica cyst; use of human feces (night soil) for soil fertilizer; contamination of foodstuffs by flies, and possibly cockroaches
- Venereal (oro-anal)

15

Clinical Classification of Amoebiasis

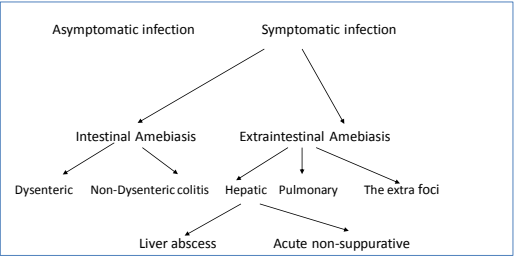
- Asymptomatic Infection: "Cyst Passers/carrier"
- Symptomatic Infection:
 - **Intestinal Amoebiasis:** (colon and rectum)
 - Acute Dysenteric (dysentery)
 - Chronic Non-Dysenteric ("self-cured")
 - **Extra-Intestinal Amoebiasis:**
 - Amoebic Liver Abscess (ALA)
 - Amoebic Pulmonary Abscess
 - Other sites (brain, skin, GU)

16



17

Clinical symptoms

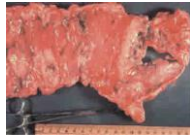


18

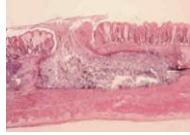
Amoebiasis: pathology

The spectrum of colitis in amoebiasis ranges from mucosal thickening, to multiple cyst formation, to diffuse inflammation / oedema, to necrosis and perforation of colonic wall.

Amoebic colitis with multiple ulcer formation



Histological cross section of classical flask shaped amoebic ulcer in colonic mucosa.



Amoebic liver abscess

- Well circumscribed abscesses are formed in the liver containing liquefied cells surrounded by inflammatory cells and trophozoites
- Adjacent parenchyma is usually unaffected



19

Amoebiasis: clinical classification

- Asymptomatic infection (carrier) >90% (*E. dispar*?)
- Symptomatic cases <10%
 - 8% -10% dysentery, colitis, etc
 - 2% invasive amoebiasis
 - 0.1% deaths

20

Acute Dysenteric Amoebiasis

Symptoms:

- Bloody mucoid diarrhea: RBCs and few WBCs in stools
- Abdominal pain; tender liver
- weight loss
- bloating, tenesmus and cramps
- Fever, 33%
- Trophozoites, but not cysts in stools
- RBCs and a few WBCs on stools

21

Chronic Non-Dysenteric Amoebiasis

"self-cured" carrier state

- Usually for 1 year, 37% symptomatic >5 years
- Intermittent diarrhea, mucus, abdominal pain, flatulence and/or weight loss
- *E. histolytica* trophozoites in loose stools
- Cysts in solid stools
- Positive serology and ulcerations on sigmoidoscopy

22

Extra-Intestinal Amoebiasis

Amoebic Liver Abscess (ALA)

- **Symptoms**
 - History of dysentery (1 yr), weight loss, abdominal pain, chest or shoulder pain
- **Signs**
 - fever, hepatomegaly
 - Diagnostic aspiration: non-odorous, reddish-brown in color aspirate (chocolate jam) "anchovy paste"
 - Might find trophozoites in the aspirate
 - Skin inflammation

23

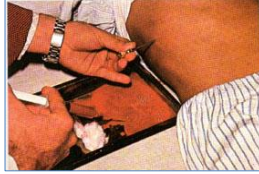


Ulcers caused by invasion of *E. histolytica* into the liver.

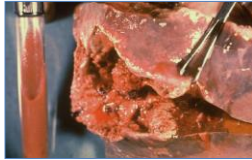
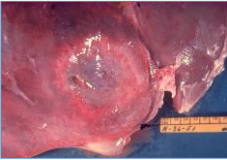
24

An Amoebic Liver Abscess Being Aspirated

Note the reddish brown color of the pus ('anchovy-sauce'). This color is due to the breakdown of liver cells.



Gross pathology of amoebic abscess of liver. Tube of 'chocolate' pus from abscess.



25

Entamoeba histolytica _ diagnosis

- Pathogenic diagnosis
 - Stool examination:
 - Direct Fecal Smear (trophozoites and cysts)
 - Fecal concentration and iodine dye techniques - (cysts) $ZnSO_4$ or formalin-ether
 - Cultivation
 - DNA detection
 - Sigmoidoscopy
- Serologic Tests (for chronic disease): ELISA, IHA (indirect hemagglutination)
- Imaging: X-ray; CT

26

Entamoeba histolytica _ diagnosis

Clinical Diagnosis:

- Sigmoidoscopic examination:

Presence of a grossly normal mucosa between the ulcers serves to differentiate amebic from bacillary dysentery (the entire mucosa being involved in bacillary dysentery).

- Hepatomegally

- C.B.C. : leukocytosis in Amebic dys. rises above 12000 per microliter, but counts may reach 16000 to 20000 per microliter.

27

***Entamoeba histolytica*_ laboratory diagnosis**

- ***Entamoeba histolytica*** must be differentiated from other intestinal protozoa including: *E. coli*, *E. hartmanni*, *E. dispar*,.....
- Differentiation is possible, but not always easy, based on morphologic characteristics of the cysts and trophozoites. The nonpathogenic ***Entamoeba dispar***, however, is morphologically identical to ***E. histolytica***, and differentiation must be based on **isoenzymatic** or **immunologic analysis**.
- Molecular methods are also useful in distinguishing between *E. histolytica* and *E. dispar*.

28

***Entamoeba histolytica*_ microscopy**

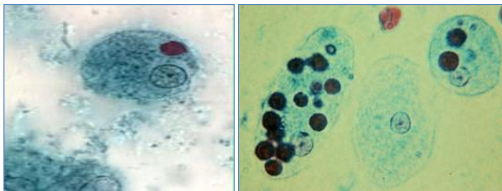
This can be accomplished using:

- Fresh stool: wet mounts and permanently stained preparations (e.g., **trichrome**).
- Concentrates from fresh stool: wet mounts, with or without iodine stain, and permanently stained preparations (e.g., **trichrome**).

29

Trophozoites of *Entamoeba histolytica*

[with ingested erythrocytes (trichrome stain)]

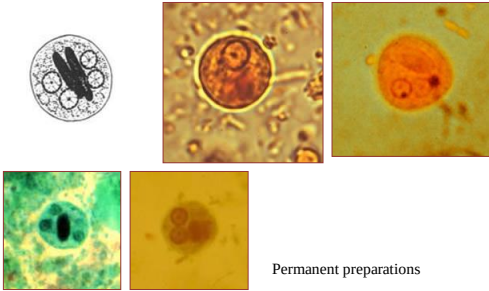


The ingested erythrocytes appear as dark inclusions.

Erythrophagocytosis is the only morphologic characteristic that can be used to differentiate *E. histolytica* from the nonpathogenic *E. dispar*.

30

Cysts of *Entamoeba histolytica*/*E. dispar*



Permanent preparations

31

Intestinal amoebiasis _ treatment

- Asymptomatic amoebiasis (cyst passer):
Diloxanide furoate (furamide): 500 mg 3 times daily / 10 days
- Symptomatic amoebiasis (troph. & cyst):
Iodoquinol , 650 mg 3 times daily/20 days
Metronidazole (Flagyl) , 750 mg 3 times daily/ 10 days
- Amebic colitis: Chloroquine, 250 mg 2 times daily
- **Acute amebic dysentery:** Emetine hydrochloride, 1mg/kg daily IM or SC

32

Extaintestinal amoebiasis _ treatment

Amebic liver abscess, amoeboma:

Metronidazole, 750 mg 3 times daily/ 10 days
plus dehydroemetine / 10 days

or

Metronidazole or dehydroemetine as above plus Chloroquine ,
500 mg 2 times daily / 2 days,.....

33

Giardia intestinalis

- Also known as:
 - Giardia lamblia*
 - Lamblia intestinalis*
 - Giardia duodenalis*
- Isolated from humans, domestic animals, and wild animals



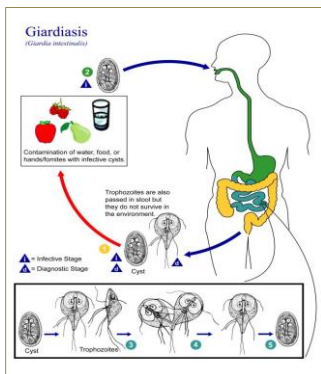
34



35

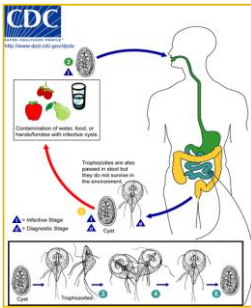
Giardia lamblia (intestinalis)

Life cycle



36

Giardia intestinalis _ Life Cycle



- Cysts responsible for transmission
- Cysts and trophozoites found in feces
- Ingested by host
- Importance of animal reservoirs

37

Giardia intestinalis _ parasite stages

- Two stages of the parasite:

Cyst



Trophozoite



38

Giardia intestinalis: transmission

- Cysts
 - Direct transmission
 - Fomites
 - Contaminated water and/or food
- Ingested cysts release trophozoites
- Trophozoites multiply and encyst in intestines
- Excreted in feces

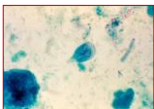
39

Giardia intestinalis

- Infects the duodenum and small intestine.
- Infection outcomes range from asymptomatic colonization to acute or chronic diarrhea and malabsorption.
- More prevalent in children

40

Giardia intestinalis: epidemiology & pathology



Intestinal flagellates

- Epidemiology - prevalence 1 to 30%, common in children's day care centers; can be transmitted in water.
- Pathology and Clinical Manifestations - symptoms can be severe; diarrhea, foul-smelling, greasy, mucus-laden stools, flatulence, nausea, cramps.
- Most infections are asymptomatic; chronic cases experience weight loss, malabsorption of fat, protein, folic acid, and fat-soluble vitamins.

41

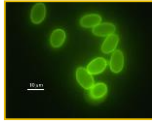
Giardia intestinalis _ disease in humans

- Illness usually lasts for 1-2 weeks
- Chronic infections reported
 - May last months to years
 - Immunodeficient and immunocompetent individuals
 - May lead to malabsorption syndromes, vitamin deficiencies, severe weight loss, and debilitation
- Disaccharide intolerance

42

Giardia intestinalis _ diagnosis

- Direct observation in feces
 - Trophozoites
 - “Tear drop” shape
 - Two nuclei and tumbling mobility
 - Cysts
 - Approximately 13 microns long
 - Oval, with 2-4 nuclei
- Immunofluorescence
- ELISA, PCR



43

Giardia intestinalis _ treatment

- Anti-protozoal drugs
- Chronic cases
 - May be resistant
 - Prolonged therapy may be necessary

44

Giardia intestinalis _ treatment

First line

- Metronidazole: 15 mg/kg/day in 3 divided doses for 5–7 days
- Tinidazole: >3 yr: 50 mg/kg/day once daily
- Nitazoxanide

Second line alternatives:

- Furazolidone 6 mg/kg/day in 4 divided doses for 10 days
- Albendazole: >6 yr: 400 mg once a day for 5 days
- paromomycin, and
- Quinacrine :6 mg/kg/day in 3 divided doses for 5 days
- Ornidazole: ??

45

46



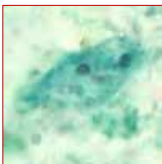
Dientamoeba fragilis



- Formerly classified as an amoeba; electron microscopy and immunological studies have suggested a flagellated cellular component.
- Laboratory diagnosis - detection of bi-nucleated trophozoites; fragmented karyosomes consisting of 4 to 8 granules of chromatin.

47

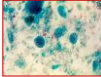
Dientamoeba fragilis



- Pathology - infection is usually asymptomatic; can be associated with diarrhea, anorexia, abdominal pain.
- Association with pinworm - the organism may be transmitted from host to host within the egg of *Enterobius vermicularis*.
- Distribution - worldwide, there is a 1% to 20% prevalence.

48

Chilomastix mesnili



- Non-pathogen
[must be differentiated from Giardia]
- Found in cecum and colon.
- Transmission - by ingestion of mature cysts.
- Trophozoite - 4 flagella (3 anterior, 1 associated with the cytostome; one nucleus, always located anteriorly).
- Cyst - lemon shape; 1 nucleus; cytostome may be seen.

49



50

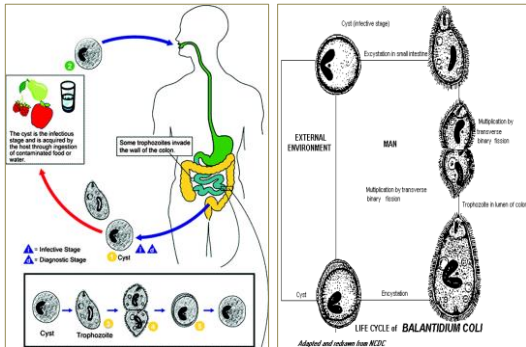
Balantidiasis

Balantidial dysentery

Ciliary dysentery

51

Balantidium coli



52

Balantidium coli

Balantidium coli has 2 developmental stages: a trophozoite stage and a cyst stage.

The cyst is the infective stage. Once the cyst is ingested via feces-contaminated food or water, it passes through the host digestive system. The tough cyst wall allows the cyst to resist degradation in the acidic environment of the stomach and the basic environment of the small intestine until it reaches the large intestine.

Excystation takes place in the large intestine; producing trophozoites from the cyst stage.

The motile trophozoite then resides in the lumen of the large intestine, feeding on intestinal bacterial flora and intestinal nutrients.

Trophozoites multiply by asexual binary fission or sexual conjugation (with the exchange of nuclear material).

The trophozoite may become invasive and penetrate the mucosa of the large intestine.

Trophozoites are released with the feces, and encyst to form new cysts. Encystation takes place in the rectum of the host as feces are dehydrated or soon after the feces have been excreted.

Cysts in the environment are then ready to infect another host.

53

Balantidium coli: epidemiology

Most infections in man are asymptomatic.

Infection in man becomes common among debilitated/ poorly nourished individuals

Animal reservoirs: pigs mainly, possibly rats too.

54

Balantidium coli: clinical features

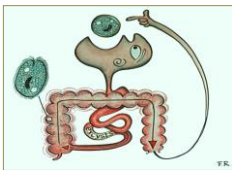
Symptomatic patients generally suffer disease similar to amebiasis, with diarrhea, dysentery, abdominal pain and weight loss.

Chronic disease is most common, although fulminant colitis may occur, including perforation leading to peritonitis.

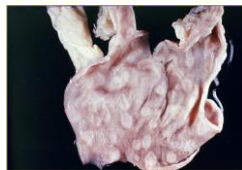
Disease in the lung, urinary bladder and bone has been described.

Similarly to *E. histolytica*, *B. coli* causes flask shaped ulcers in the large intestine, most commonly in the cecum and rectosigmoid

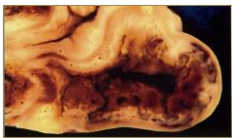
55



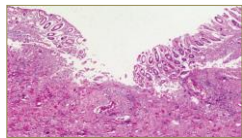
Ingested cysts excyst in the intestine to become trophozoites.



Numerous ulcers in colon of patient with balantidiasis.



Necrotic lesion of large intestine caused by *Balantidium coli*.



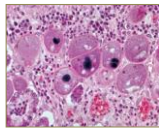
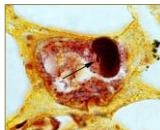
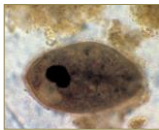
Flask-shaped ulcer in colon of patient with balantidiasis.

56

Balantidium coli: diagnosis

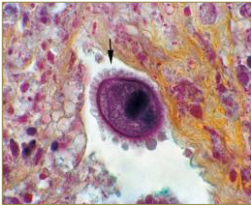
Diagnosis is usually made by identification of mobile trophozoites in fresh stool or scraped from an ulcer seen during endoscopy.

- Trophozoite is oval, large (some times can be seen without magnification)
- Contain circumferential cilia; macro & micro nuclei.
- In tissue sections, large kidney/bean shaped nuclei, micro-nucleus and cilia are visible.
 - Precysts are round, ciliated, and have a peristome. Cysts, which are not observed in histologic sections of colon, are spherical and vary from 45 μm to 75 μm in diameter.
 - Rarely the diagnosis is made by examination of urine or bronchoalveolar lavage fluid, or by identification in biopsy or resection specimens.



57

Balantidium coli: trophozoites in tissue



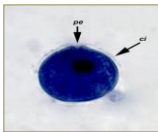
Balantidium coli trophozoite in colon. Note prominent cilia (arrow), cell membrane, and large macronucleus



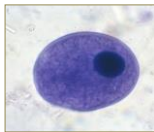
Balantidium coli in colonic ulcer. Note cilia-lined peristome (arrow) at anterior end

58

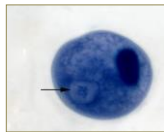
Balantidium coli: cysts in colon and stool



Balantidium coli trophozoite in colon demonstrating kidney-shaped macronucleus. I & H



Balantidium coli cyst in stained stool specimen illustrating cyst wall and single large macronucleus. I & H



Balantidium coli cyst in stained stool specimen showing single large macronucleus and food vacuole (arrow) containing bacteria. I & H

59

Balantidium coli: Treatment

Tetracyclines: 500mg 4x/daily, 10 days

Metronidazole: 750 mg 3x/daily, 5 days

Iodoquinol: 640 mg 3x/daily, 20 days

Others: Nitazoxanide, Ampicillin

60

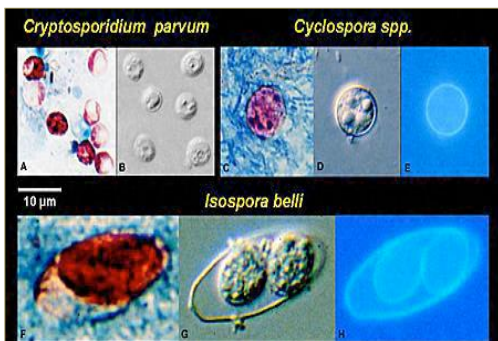


61

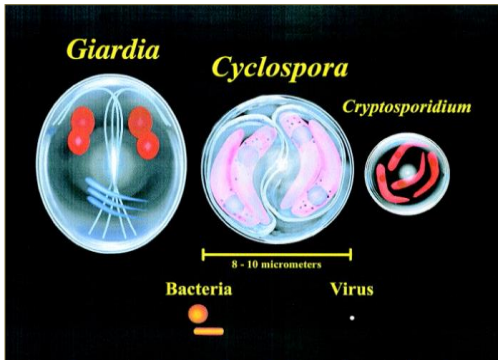
Coccidian parasites

- Organisms infecting humans include:
Isospora, *Sarcocystis*, *Cryptosporidium*, *Cyclospora* & *Toxoplasma*.
- Some have 2-host life cycle.
- Sexual & asexual reproduction - Schizogony - asexual binary fission; Sporogony - sexual reproduction.
- Diagnostic stages are often difficult to locate. Acid fast stains are used to visualize. Oocysts do not stain with iodine or permanent stains like trichrome.

62



63



64

Cryptosporidium

- 4-5 μm oocysts
- no sporocysts
- 4 sporozoites

Cyclospora

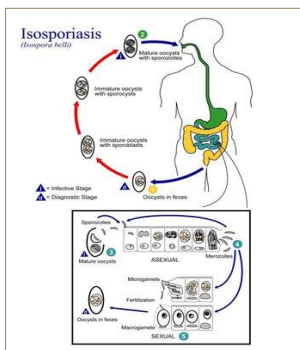
- 8-10 μm oocysts
- 2 sporocysts
- 2 sporozoites each

Isospora belli

- 30 x 12 μm oocysts
- 2 sporocysts
- 4 sporozoites each

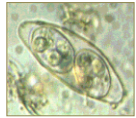
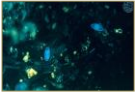
65

Isosporiasis: life cycle



66

Isospora belli



- Diagnostic/infective stage - mature oocyst - contains two sporocysts, each of which contain four sporozoites (do not stain with iodine).
- Intermediate hosts - none.
- Definitive host: humans
- Pathology - ranges from asymptomatic to acute, severe dysentery.
- The least common among coccidian parasites

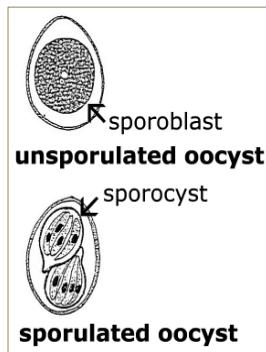
67

Isospora belli



- Schizogony - takes place upon initial infection when sporozoites invade the intestinal epithelium and multiply.
- Sporogony - progeny of asexual reproduction initiate development into gametes (male & female); fertilized gametes develop into oocysts which are passed as the infectious stage in the feces.

68



69

Isosporiasis: pathogenesis

Incubation period: 3-14 days.

Symptoms:

- Profuse, watery, nonbloody, offensively smelling diarrhea, which may contain mucus
- Cramping abdominal pain, vomiting
- Malaise, anorexia, weight loss
- Low-grade fever
- Steatorrhea in protracted cases

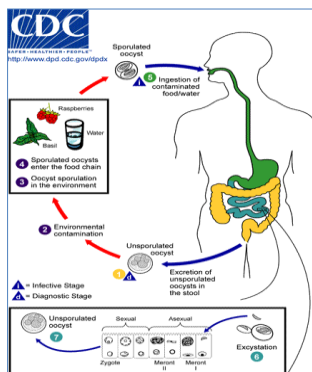
70

Isosporiasis: pathogenesis

Trimethoprim-sulfamethoxazole (TMP-SMZ) is the drug of choice because it is the best-studied and most readily available agent.

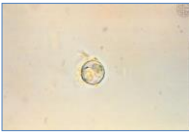
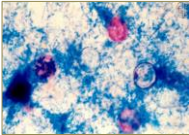
71

Cyclospora cayetanensis



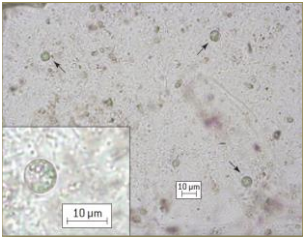
72

Cyclospora cayetanensis

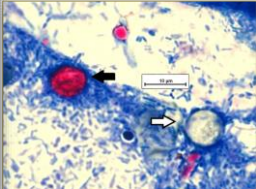
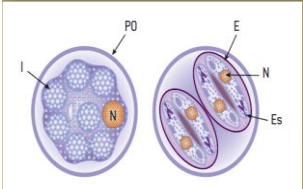


- *C. cayetanensis* is the name designated for the only one known to infect humans.
- Animal reservoirs - not known.
- Pathology - diarrhea similar to that experienced with *Cryptosporidium* infections; pathology in intestines like *Isospora*.
- Identification - Oocysts are 10 microns in diameter, and are variably acid-fast.

73



Cyclospora cayetanensis



74

Cyclospora cayetanensis

Sporulation: cyst, sporocyst, sporozoites
Cyclospora oocyst (two sporocysts with two sporozoites each)

Disease: Diarrhea (among travelers and immune compromised)
5 - 10% of diarrheal episodes during rainy seasons

Transmission: Food and water (Raspberry consumption)
There are data suggesting that lettuce, chicken, and unpasteurized milk are other potential sources of infection
Oocysts stay moist on fresh fruit and vegetables at 22°C - 32°C.



Guatemalan raspberries

75

Cyclospora cayetanensis

Clinical manifestations

Infection:

Self-resolving in immune competent persons

Organism can complete its life cycle within a single human host

Duration of illness:

A few days/3 months (average 6-7 weeks)

Chronic symptoms:

A syndrome of prolonged, intermittent diarrhea associated with profound fatigue and anorexia.

Severe symptoms:

Abrupt onset of watery diarrhea, fever, nausea, vomiting, and abdominal cramps (30% of cases)
Resolve in 2-3 days. Fatigue and loss of appetite.

76

Cyclospora cayetanensis

Pathogenesis

Cyclospora infects the upper intestine (duodenum, jejunum)

▪ Flattening of the villi.

▪ Inflammatory infiltrates of the basement membrane.

N.B. Pathology of Cyclospora in the intestine is similar to that of Isospora

77

Cyclospora cayetanensis

Laboratory diagnosis

Microscopic identification of oocysts in stool specimens

Treatment

Drug of choice for treatment:

Trimethoprim-sulfamethoxazole (TMP-SMX)

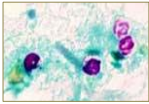
Dose: (2x/daily, 7 days)

160 mg trimethoprim, 800 mg sulfamethoxazole

N.B. = Cyclospora and Isospora infections are similar with regard to their susceptibility to TMP-SMX in vivo.

78

Cryptosporidium parvum



- Site of infection - primarily an intestinal infection, can become systemic in AIDS patients.
- Originally considered an animal parasite (rodents, cattle and sheep).
- Infections in immunocompromised people can be serious, even fatal.
- Identification - oocysts are 2 - 5 microns in diameter; do not stain with iodine; and are acid-fast.

79

Cryptosporidium parvum

Life cycle

The life cycle begins with the ingestion of oocysts by the host.

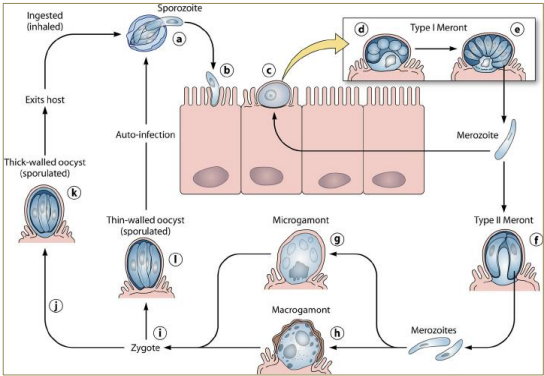
Following excystation, four naked sporozoites are released in the gut, which then infect epithelial cells and initiate asexual development. Sporozoites become internalized and undergo two successive generations of merogony, releasing eight and four merozoites, respectively.

The four merozoites released from the second merogony give rise to the sexual developmental stages, the micro- and macrogametes.

The release of microgametes, and their union with macrogametes, gives rise to the zygote, which, after two asexual divisions, forms the environmentally resistant oocyst containing four sporozoites.

80

Cryptosporidium parvum: life cycle



81

Cryptosporidium parvum

Transmission

Transmission is fecal-oral, either directly or indirectly via contaminated water or food washed or irrigated with fecally contaminated water.

Human and dairy effluents are probably important sources of infection

Oocysts are environmentally highly resistant to common disinfectants; they are discharged in large numbers by infected humans and animals, and via effluent disposal, find their way to surface water.

C. parvum oocysts are released in large quantities from clinically infected humans and calves (> 10¹⁰ during acute or chronic infections), and less from asymptotically infected individuals

82

Cryptosporidium parvum



- Transmission - the endemic cycle is maintained fecal-oral route transmission; a relatively common finding in "day-care" diarrhea;
- Big potential for being waterborne due to significant resistance to disinfectants.
- Pathology - most infections cause severe diarrhea.
- In the immunosuppressed patient, the condition is protracted and life threatening.
- There is no drug effective against this parasite.

83

Cryptosporidium parvum

Pathogenesis & disease

C. parvum causes an acute, self-limiting infection and diarrheal disease in immunocompetent people, in whom the onset may be rapid (3–7 d):

Variability of onset depends on:

- Age
- Presence of maternal antibodies or previous exposure
- Infectious dose (10 or less oocysts can be infective)
- Parasite (origin and age of oocysts, and species/genotype) factors.

84

Cryptosporidium parvum

Pathogenesis/disease

Infection begins in the small intestine, where the emerging sporozoites infect enterocytes, and after amplification, endogenous forms spread throughout the epithelial surfaces of both villi and crypts.

The infection may spread throughout the gut, which includes the gastric mucosa and the small and the large intestines, or it may remain localized in segments of the small and/or large intestine.

The extent of spread and the sites involved determine whether the infection is clinical or subclinical as well as the overall intensity of the disease.

85

Cryptosporidium parvum

Pathogenesis & disease in immune compromised individuals

Cryptosporidiosis is considered to be one of the **most serious opportunistic infections** that complicates AIDS.

Individuals with CD4 T-cell counts of <150/ml who become exposed to *C. parvum* *invariably develop persistent infection*, with profound and life-threatening diarrhea.

Prolonged infections lasting several months or years in people with acquired or congenital immunodeficiencies often spread from the gut to the hepatobiliary and the pancreatic ducts, causing cholangiohepatitis, cholecystitis, choledochitis or pancreatitis.

In chronically infected gut, the mucosal architecture undergoes gradual but profound disorganization, which includes disrupted epithelial surface, fibrosis, cellular infiltration and crypt abscessation.

86

Cryptosporidiosis: symptoms

- Incubation period: 1 to 12 days
- Clinical signs
 - Profuse, watery diarrhea
 - Abdominal pains
 - Anorexia
- Self-limiting in healthy people
- May become serious in immunocompromised people

87

Cryptosporidium parvum: diagnosis

Detection depends on the presence of intact oocysts in feces. Because clinically affected individuals excrete them in abundance, the diagnosis of *C. parvum* is not difficult.

The earliest method of staining of a fecal smear with modified acid fast (MAF) remains the quickest (~15 min) and the easiest to perform if the sample is reasonably fresh.

Oocysts measure 3–5 µm, are round and stain bright red, surrounded by a refractile rim, and often, 2–4 sporozoite nuclei can be observed.

Other methods used in clinical diagnostic laboratories include direct or indirect immunofluorescence microscopy and ELISA.

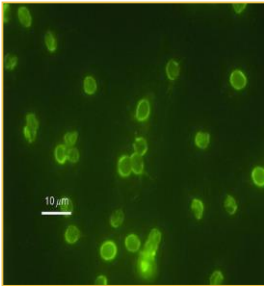
None of these techniques allows discrimination regarding the species of origin of the oocysts, or whether they are infectious or not.

Only molecular techniques, including PCR, used in most investigative laboratories, can help determine the possible source and risk to human health.

88

Cryptosporidium parvum: Diagnosis

- Differentials
 - *Giardia*, *Iso*spora
 - Microsporidia
 - *Salmonella*
 - Rotavirus
- Diagnostic tests
 - Fecal flotation
 - Acid-fast staining
 - Immunofluorescence



89

Cryptosporidium parvum

Treatment

Non-specific supportive treatment, including rehydration and nutritional supplementation, remains a mainstay of management of the clinical manifestations of cryptosporidiosis.

In AIDS patients, reduction in viral load and concomitant rise in CD4 counts achieved by antiretroviral therapy results in rapid clinical improvement in symptoms as well as a reduction in oocyst excretion.

Modest treatment benefits from: paromomycin or nitazoxanide
Paromomycin (may be used in combination with azithromycin)

90



91

Diphyllobothrium latum

The cestode *Diphyllobothrium latum* (the fish or broad tapeworm), the largest human tapeworm.

92

D. latum containing many proglottids



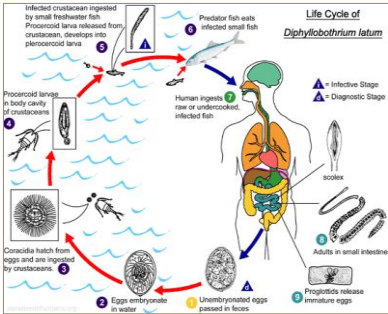
Scolex of *D. latum*



93

Diphyllobothriasis

Life cycle



94

Diphyllobothriasis

Infection, transmission

Infection with the adult worm is acquired by the ingestion of raw, poorly cooked or pickled fresh water fish containing the encysted proceroid larvae.

After ingestion, the worm matures, with egg production beginning about the fifth or sixth week.

The adult worm reaches a length of 10m or more and may contain up to 3000 proglottids.

95

Diphyllobothriasis

Clinical disease

Diphyllobothriasis can be a long-lasting infection (decades).

Most infections are asymptomatic.

Manifestations may include abdominal discomfort, diarrhea, vomiting, and weight loss. Vitamin B12 deficiency with pernicious anemia may occur when worm attaches to jejunum

Massive infections may result in intestinal obstruction. Migration of proglottids can cause cholecystitis or cholangitis.

Symptoms in the patient depend on a number of variables: the number of worms present; the amounts and types of by-products produced by the worm; the patient's reaction to such byproducts; and the absorption of various metabolites by the worms.

96

Diphyllobothriasis

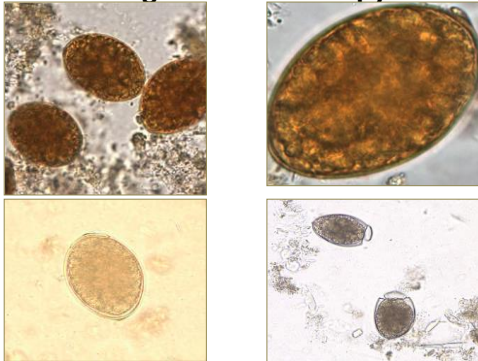
Diagnosis

Diagnosis is usually based on the recovery and identification of the characteristic eggs and/or proglottids. If the egg operculum is difficult to see, the coverslip of the wet preparation can be tapped and the pressure may cause the operculum to pop open, thus making it more visible. The eggs are unembryonated at the time they are passed in the stool.

Proglottids are often passed in chains and this is a clue to *D. latum*. The overall proglottid morphology with the rosette uterine structure also facilitates identification.

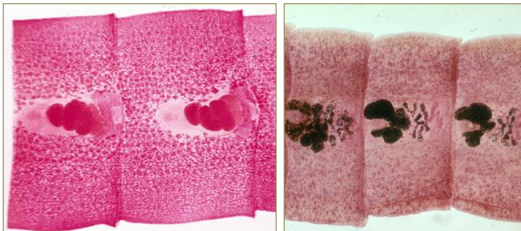
97

Diagnosis: microscopy



98

Carmine-stained proglottids of *D. latum*, showing the rosette-shaped ovaries



99

Diphylobothrium latum: treatment & prevention

Treatment:

Drug of choice: Praziquantel

2nd line: Niclosamide

Supplemental: Vitamin B12

Prevention:
Cook fish at 60°C, 5 minutes (avoid raw/undercooked fish)

Freeze at -20°C, 60 hours
